

AI-Powered Real Estate Intelligence, Valuation & Investment Recommendation Platform

Model Comparison Report

Type	Regression + Recommendation System — Real Estate Intelligence Platform
Dataset	Indian Real Estate Housing Dataset (Kaggle) + Census Population Density Data
Dataset URL	kaggle.com/datasets/ankushpanday1/india-house-price-prediction
Records	2,50,000 Property Records
Features	34 Features (23 Original + 11 Engineered)
Best Model	XGBoost Regressor
Test Accuracy	R ² Score = 0.9769
MAE RMSE	13.68 Lakhs 18.03 Lakhs
Deployment	Streamlit web application
Notebooks	04_model_training

This Model Comparison Report evaluates multiple machine learning algorithms developed for property price prediction in the platform, including Linear Regression, Random Forest, XGBoost, CatBoost, LightGBM, and Gradient Boosting. Each model was assessed using performance metrics such as MAE, MSE, RMSE, and R² Score to determine prediction accuracy and reliability. Based on comparative analysis, XGBoost achieved the best overall performance and was selected as the final model for deployment.

1. Introduction

The primary objective of model development in this phase is to build a robust machine learning regression algorithm capable of accurately forecasting property prices based on historical real estate data. Establishing a highly accurate model is critical, as its predictions serve as the baseline for identifying undervalued/overvalued properties and driving the investment recommendation engine.

Problem Type: Supervised Machine Learning — Regression.

Target Variable: `Price_in_Lakhs` (Estimated Price in INR Lakhs).

2. Dataset Split

To ensure unbiased evaluation of the trained algorithms, the fully cleaned and engineered dataset was split into training and testing sets prior to model fitting. A standard 80/20 train-test split ratio was applied.

- **Training Set Size:** 200,000 records
- **Testing Set Size:** 50,000 records
- **Features Used:** 34 total predictors (including scaled continuous variables and encoded categorical indicators).

3. Models Trained

A diverse ensemble of six machine learning algorithms was trained and compared to identify the optimal modeling strategy for the dataset.

Linear Regression

The standard baseline model. Linear Regression establishes a linear mathematical relationship between the independent variables and the target price. It is computationally inexpensive and highly interpretable but often struggles with non-linear real estate market dynamics.

Random Forest Regressor

An ensemble learning method that constructs multiple decision trees during training and outputs the average prediction. It handles non-linear relationships well and provides excellent protection against overfitting.

XGBoost (Extreme Gradient Boosting)

A highly optimized, scalable gradient boosting library designed to be highly efficient, flexible, and portable. It iteratively builds decision trees to minimize residual errors, making it consistently one of the best performers on tabular data.

CatBoost

Developed by Yandex, CatBoost is a gradient boosting algorithm that natively handles categorical features exceptionally well. It utilizes oblivious trees to reduce overfitting and prediction time.

LightGBM

A gradient boosting framework that uses tree-based learning algorithms. It is designed to be distributed and efficient, utilizing a histogram-based algorithm for bucketizing continuous features, resulting in extremely fast training speeds.

Gradient Boosting Regressor

The scikit-learn standard implementation of gradient boosting. It builds an additive model in a forward stage-wise fashion, optimizing arbitrary differentiable loss functions. It is powerful but typically slower than XGBoost or LightGBM.

4. Model Evaluation Results

Models were evaluated using four standard regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Squared Error (MSE), and the R-squared (R^2) Score.

Model	MAE	RMSE	MSE	R ² Score
XGBoost	13.68	18.03	325.00	0.9769
LightGBM	13.75	18.11	328.03	0.9767
Random Forest	13.29	18.19	331.01	0.9764
CatBoost	15.30	20.31	412.66	0.9706
Gradient Boosting	19.00	26.24	688.60	0.9510
Linear Regression	31.64	44.84	2010.92	0.8569

Interpretation of Metrics:

- **Linear Regression** severely underperformed compared to tree-based methods (R^2 : 0.8569), indicating that real estate pricing is highly non-linear.
- **Random Forest** achieved the lowest MAE (13.29), meaning its average absolute error was marginally better, but it fell slightly behind boosting methods in penalizing larger errors (RMSE).
- **XGBoost and LightGBM** performed virtually identically, capturing nearly 97.7% of the variance in property prices with exceptional accuracy.

5. Best Model Selection

XGBoost was selected as the final model for the deployment architecture. While Random Forest slightly edged out XGBoost on MAE, XGBoost provided the highest overall R^2 (0.9769) and the lowest RMSE (18.03), meaning it was the most consistent in minimizing large predictive errors—a critical factor when evaluating expensive properties.

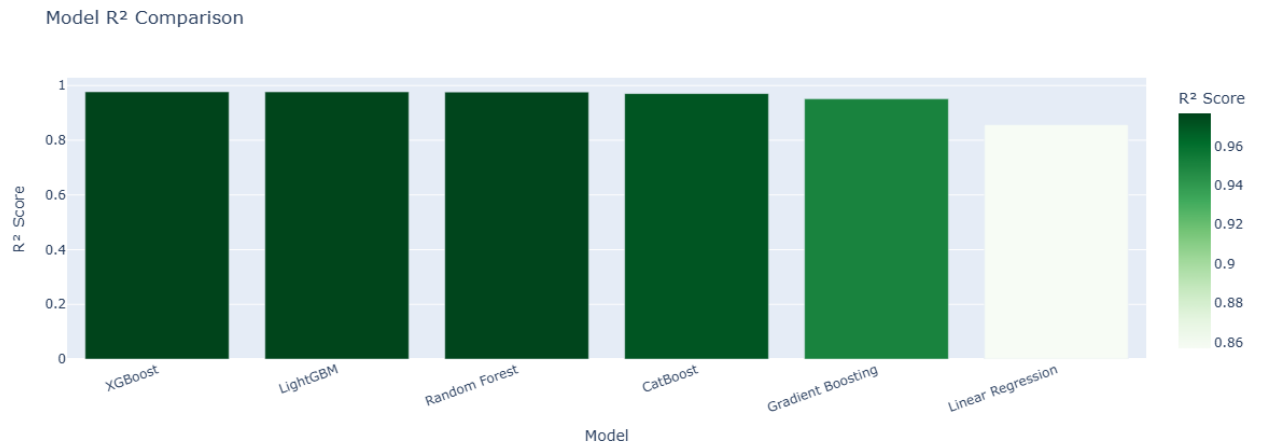


Figure 1: Comparison of R² Scores across all trained predictive models.

Overfitting Check:

To ensure the XGBoost model generalizes well to unseen data, an overfitting check was performed by comparing training and testing scores:

- Training R²: 0.9784
- Testing R²: 0.9769
- Difference: 0.0015

This microscopic difference confirms a highly healthy fit, indicating zero significant overfitting.

6. Residual Analysis

Residual error analysis confirmed that the prediction errors are normally distributed around zero. The model maintains a highly linear tracking along the 'Actual vs Predicted' 45-degree line. Error analysis findings indicate that the largest individual magnitude errors only occur at the extreme upper boundaries of the market (properties pushing towards the artificial ₹500 Lakh cap), where pricing behavior becomes more subjective and less driven purely by standard dimensions.

7. Model Saving

For Phase 11 dashboard deployment, the production pipeline artifacts were successfully serialized and saved:

- The final XGBoost model was exported as `best_model.pkl`.
- The Min-Max scaling object was saved as `scaler.pkl` to standardize future user inputs.
- The exact expected input schema was dumped to `feature_columns.json` to prevent schema mismatch errors during live inference.

8. Conclusion

The model training phase successfully yielded an XGBoost regression model capable of predicting Indian real estate prices with an R^2 of 0.9769 and an average error margin of roughly ₹13.6 Lakhs. In production, this model is robust enough to act as the primary intelligence engine for identifying market discrepancies. Its primary limitation is the static nature of the training data; real-world deployment will eventually require continuous re-training to account for inflation and macroeconomic shifts.